Structured Multi-Evidence Learning for Hyperspectral Image Clustering

Huilang Xu, Kaiyu Lin, Pan Gao, Guang-Yong Chen, *Member, IEEE*, Xing Chen, *Member, IEEE*, Min Gan, *Senior Member, IEEE*, and C. L. Philip Chen, *Life Fellow, IEEE*

***Abstract*—The hyperspectral images (HSIs) record subtle spectral responses of land-cover materials over a large number of contiguous bands, providing rich information for unsupervised land-cover analysis. However, local environmental variations, spectral ambiguity, and band redundancy often make the underlying clustering structure unreliable. The existing HSI clustering methods usually start from a single input evidence source, such as spectral features or current embeddings, and then improve clustering stability through structural priors, graph relations, pseudo labels, or augmented views. Although useful, these signals are still largely derived from the same base evidence source and may inherit its bias when the induced clustering geometry is unreliable. To address this issue, we propose a structured multi-evidence learning (SMEL) framework for lightweight hyperspectral clustering. Instead of merely deriving constraints or perturbed views from the same base evidence source, the proposed method explicitly organizes complementary evidence from spectral characteristics, region context, and region prototype relations before assignment learning, allowing different spectral–spatial cues to compensate for each other. Benefiting from this structured evidence organization, the proposed lightweight multi-evidence assignment learning module establishes semantic consensus among complementary evidence sources without relying on complex deep encoders or elaborate handcrafted priors. Experiments on several benchmark hyperspectral datasets show that the proposed structured multi-evidence framework achieves stable and competitive clustering performance with only a few thousand learnable parameters. Our code is available at https://github.com/NuclearLemon/SMEL.**

***Index Terms*—Hyperspectral image (HSI) clustering, lightweight clustering, multi-evidence learning, structured evidence organization.**

Received 25 June 2026; revised 9 August 2026; accepted 2 September 2026. Date of publication 7 September 2026; date of current version 25 September 2026. This work was supported in part by the "Open Competition" Project of Xiangjiang Laboratory under Grant 25XJJB02; in part by the Foundation for Outstanding Research Groups of the National Natural Science Foundation of China under Grant 72688201; in part by the National Key Research and Development Program of China under Grant 2025YFB4507104; in part by the National Natural Science Foundation of China under Grant 62173091; in part by the Natural Science Foundation of Fujian Province under Grant 2024J09021; and in part by the Postgraduate Education Teaching Reform Project of Fuzhou University under Grant FYAI2025011. *(Corresponding author: Guang-Yong Chen.)*

Huilang Xu, Kaiyu Lin, Pan Gao, Guang-Yong Chen, and Xing Chen are with the College of Computer and Data Science, Fuzhou University, Fuzhou 350108, China, and also with Fujian Key Laboratory of Network Computing and Intelligent Information Processing, the Key Laboratory of Intelligent Metro of Universities in Fujian, and the Engineering Research Center of Big Data Intelligence, Ministry of Education, Fuzhou 350108, China (e-mail: xuhuilang1999@outlook.com; 2501020033@fzu.edu.cn; 241020026@fzu.edu.cn; cgykeda@mail.ustc.edu.cn; chenxing@fzu.edu.cn).

Min Gan is with the Xiangjiang Laboratory, Changsha 410205, China (e-mail: ganmin@ieee.org).

C. L. Philip Chen is with the School of Computer Science and Engineering, South China University of Technology, Guangzhou 510006, China (e-mail: Philip.Chen@ieee.org).

Digital Object Identifier 10.1109/TGRS.2026.3731375

I. Introduction

THE hyperspectral images (HSIs) record contiguous spectral bands that characterize materials and land cover, supporting tasks, such as classification and reconstruction [1], [2]. Since task-specific supervision is often costly or unavailable, the unsupervised HSI analysis has received increasing attention [3]. The HSI clustering partitions pixels or regions into semantically meaningful land-cover categories without manual annotations, with applications in land-cover mapping, environmental monitoring, and resource exploration [4], [5].

However, due to illumination changes, sensor noise, and local spatial environmental variations, the pixels from the same land-cover class may exhibit significantly different spectral responses, while the pixels from different classes may show highly similar spectral curves [6]. Meanwhile, the adjacent spectral bands are often highly redundant, and some bands may contain weakly discriminative or noisy information [7]. Under the single-scene unsupervised learning setting, class imbalance, small objects, mixed boundaries, and intraclass multimodality further increase the uncertainty of the clustering structure [8]. Therefore, the central challenge of hyperspectral clustering lies in how to construct reliable clustering evidence and obtain stable assignments from complex, redundant, and scene-dependent spectral–spatial data.

To address this challenge, the existing hyperspectral clustering methods have mainly developed along two lines. The classical optimization-based methods usually formulate clustering with explicit structural assumptions, such as subspace models [9], low-rank representations [10], and graph relations [11]. These methods often have interpretable objective functions and well-defined structural priors. More recently, the deep learning methods have introduced convolutional networks [12], transformers [13], and contrastive learning [14] to improve nonlinear representation ability. These studies have significantly advanced unsupervised hyperspectral clustering by introducing structural constraints, graph propagation, pseudo labels, or augmented views on top of the input representation or current embedding.

Despite these advances, the reliability of clustering evidence remains a fundamental issue in HSI clustering. Due to spectral ambiguity, band redundancy, and local spatial variations, different descriptions of the same HSI may induce biased clustering geometries. The raw spectra preserve complete band responses, but they are sensitive to high-dimensional redundancy and noisy bands. PCA reduces dimensionality and retains major global variance, but its objective is not directly related to class separability and may weaken discriminative local spectral variations [15]. Superpixel-wise regional

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

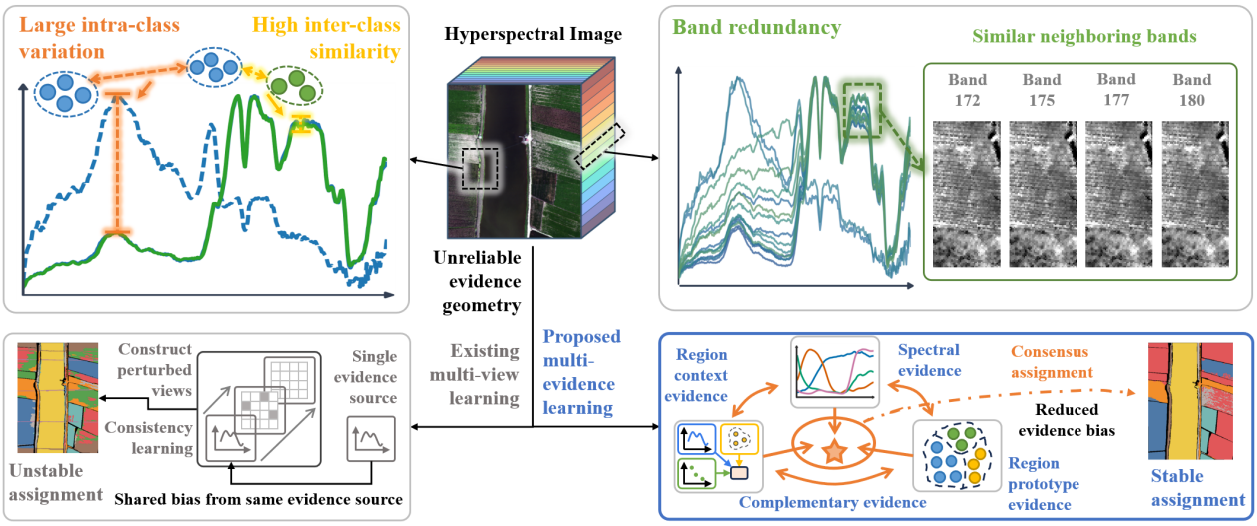


Fig. 1. Motivation of the proposed SMEL framework. HSIs often suffer from large intraclass variation, high interclass similarity, and band redundancy, which can make the evidence geometry unreliable for clustering. Existing methods usually rely on a single base evidence source; even with additional priors or augmented views, the resulting self-supervised signals may still inherit its bias. In contrast, the proposed framework organizes evidence from spectral characteristics, region context, and region prototype relations to make complementary spectral–spatial cues compensate for each other; furthermore, a lightweight multi-evidence assignment learning module is introduced to establish semantic consensus in a unified assignment space, reducing the dependence on biased single-source evidence under uncertain spectral–spatial structures.

averages introduce local spatial consistency [16], but they may also smooth out pixel-level differences around mixed or inaccurate region boundaries. As conceptually shown in Fig. 1, large intraclass variation, high interclass similarity, and band redundancy can make the evidence geometry ambiguous. However, many existing methods still first form a base evidence source from raw spectra, PCA-reduced features, or current embeddings and then derive self-supervised clustering signals from this base evidence source. For example, graph edges are usually estimated from spectral–spatial similarity, pseudo labels are generated from initial clustering results, and multiview contrastive signals are often produced by perturbing or sampling the same evidence source [17]. These strategies are useful for improving robustness. Nevertheless, when the base evidence source is unreliable, the derived graph relations, pseudo labels, or perturbed views may inherit similar bias and propagate it during training. Therefore, an important issue is not only how to design stronger encoders or add more constraints after an input representation has been formed but also how to organize complementary evidence sources before assignment learning, so that different spectral–spatial cues can compensate for each other in a unified clustering space.

Motivated by this observation, we propose a structured multi-evidence learning (SMEL) framework for HSI clustering. Different from common multiview strategies that generate perturbed views from the same base evidence source, our goal is to construct complementary spectral–spatial evidence before assignment learning. Specifically, we organize the same HSI into compact evidence sources that describe spectral characteristics, region context, and region prototype relations. These evidence sources provide different but related cues for the same clustering task, reducing the risk that assignment learning is dominated by one unreliable evidence source. A lightweight low-rank encoder extracts evidence-specific representations, and a shared assignment head maps them into a unified assignment space. In this way, different evidence sources can support assignment-level consensus while preserving their own discriminative information. Experimental results show that the proposed framework achieves competitive clustering performance with a compact assignment model. The main contributions of this article are summarized as follows.

1) *SMEL Framework:* We propose a SMEL framework for HSI clustering. To alleviate the unreliability of a single evidence source caused by spectral ambiguity and band redundancy, the proposed framework no longer merely derives constraints or perturbed views from the same base evidence source. Instead, it organizes evidence from spectral characteristics, region context, and region prototype relations before assignment learning, thereby reducing the dependence of clustering assignments on one unreliable evidence source.
2) *Lightweight Multi-Evidence Assignment Learning:* Benefiting from structured multi-evidence organization, the model does not rely on complex deep encoders or elaborate handcrafted priors to compensate for the limitations of a single evidence source. Instead, it captures evidence-specific information and establishes semantic consensus among complementary evidence sources with only a small number of learnable parameters.
3) *Systematic Experimental Validation:* We conduct systematic experiments on multiple benchmark hyperspectral datasets. The results show that the proposed method achieves stable and competitive clustering performance with only a few thousand learnable parameters.

The remainder of this article is organized as follows. Section II reviews the related work on HSI clustering. Section III presents the proposed SMEL framework. Section IV reports the experimental results and analyses. Finally, Section V concludes this article.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

A. Classical and Deep HSI Clustering

The HSIs have been widely used in classification [18], [19], [20] and reconstruction [21]. When reliable annotations or paired reference data are unavailable, HSI clustering provides an unsupervised means of discovering intrinsic land-cover structures [22]. The existing HSI clustering methods can be roughly grouped into classical optimization-based methods and deep learning-based methods. Classical methods usually formulate clustering as an explicit optimization problem by exploiting subspace structures, graph relations, or anchor-based approximations [23]. For example, the spectral–spatial sparse subspace clustering incorporates spatial information into sparse subspace modeling [24], while anchor graph and bipartite graph methods reduce the cost of full graph construction and improve scalability for large hyperspectral scenes [11], [25]. Recent projected or structured anchor learning methods further enhance the efficiency of large-scale clustering by jointly learning projections and representative anchors [9], [26]. However, their effectiveness is often tied to the similarity geometry induced by the input representation, predefined graph construction rules, or structural assumptions. When the input evidence is affected by band redundancy, noise, or spectral ambiguity, the subsequent optimization process may still be constrained by a biased evidence geometry.

Deep learning methods have been introduced to improve nonlinear representation ability for hyperspectral clustering [13]. These methods can capture flexible nonlinear representations [27], [28], but their self-supervised signals are commonly constructed from current features or derived structures, such as reconstruction consistency, graph neighborhoods, pseudo labels, or contrastive pairs [29], [30]. These signals are useful for representation learning, yet they are often derived after a base input representation or evidence source has been formed. Therefore, if the base evidence source is unreliable, the learned representation may inherit or amplify its bias during training. In addition, heavy encoders may introduce higher computational cost and weaker interpretability in the single-scene unsupervised setting. These observations motivate a complementary direction: improving not only the representation model but also the construction and organization of clustering evidence before assignment learning.

B. Spatial Structural Priors

Since hyperspectral clustering is performed without manual annotations, constructing reliable self-supervised signals is crucial. Spatial structural priors are therefore widely used in this task [31]. Among them, superpixels are commonly adopted to group pixels into spatially coherent regions, which can reduce computational cost and provide local consistency constraints. The existing methods use superpixels as local aggregation units, region prototypes, positive sample sets, or graph nodes to suppress pixel-level noise and stabilize clustering [32], [33]. When the partition is reliable, the pixels within the same superpixel tend to share similar semantics, making superpixels useful local priors for unsupervised clustering.

Graph-based structures further model spatial relations among pixels or regions. By constructing pixel graphs, superpixel graphs, anchor graphs, bipartite graphs, or hypergraphs, these methods propagate spectral–spatial similarities and improve the spatial continuity of clustering assignments [11], [14], [25]. However, the reliability of graph relations depends strongly on the initial features, similarity metrics, and regional partitions [34]. Under spectral ambiguity and high interclass similarity, inaccurate graph edges may connect samples from different classes, causing wrong relations to be propagated through graph smoothing, graph propagation, or graph contrastive learning [12]. To alleviate this issue, recent methods introduce adaptive edge learning to refine graph structures during representation learning [35]. These studies indicate that structural reliability is a key issue in hyperspectral clustering.

Nevertheless, spatial priors are often introduced as constraints, graph relations, or training signals after a base representation has been obtained. When the initial representation is severely confused, it remains difficult to reliably distinguish wrong edges, true boundary edges, and the same-class connections across distant regions. Moreover, local superpixel consistency can improve regional smoothness and stability, but it is not sufficient to establish a unified clustering coordinate for spatially separated regions belonging to the same semantic class. In this work, superpixel information is therefore not treated only as a post-hoc smoothing or graph construction tool. Instead, region context and region prototype relations are incorporated as complementary evidence sources, together with spectral characteristics, to support unified assignment learning.

C. Multiview Learning

Multiview learning has also been explored in HSI clustering [36]. In multisource remote sensing scenarios, different views are usually provided by different sensors or modalities, such as hyperspectral imagery, light detection and ranging, synthetic aperture radar, or other auxiliary observations [37]. The main objective of these methods is to fuse complementary information across heterogeneous data sources [38]. This work, instead, considers the single-source hyperspectral clustering setting, where all clustering evidence is derived from different descriptions of the same HSI. Therefore, the key issue is not cross-modal fusion, but how to construct and organize complementary spectral–spatial evidence from a single HSI.

Recent hyperspectral clustering methods also construct augmented or contrastive views through spectral–spatial perturbation, graph perturbation, noise injection, or data augmentation and then use cross-view consistency or contrastive learning to obtain invariant representations [39], [40]. These methods show that view diversity can provide useful self-supervised signals. However, these views are typically generated by perturbing or augmenting a base representation to improve invariance or robustness. In this work, we aim to construct complementary clustering evidence from a single HSI. These evidence sources are preserved as separate branches and fused in a common assignment space, rather than being used mainly as augmented views for feature-level invariance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

Most closely related to our setting, S^{2}ESCC [41] constructs two spectral subviews from a single HSI and learns superpixel-level graph embeddings with a dual-branch graph neural network and contrastive losses. Different from S^{2}ESCC, our focus is not to design a specific spectral split followed by graph contrastive learning but to organize complementary evidence from spectral characteristics, region context, and region prototype relations for assignment-level consensus learning.

III. Method

A. Overview

As discussed above, a key difficulty in HSI clustering is not only how to learn stronger representations but also how to organize reliable clustering evidence under spectral ambiguity, band redundancy, and spatial mixing. Existing methods often derive graph relations, pseudo labels, or perturbed views from a formed input representation. When this base evidence source is biased, subsequent constraints or augmentations may still inherit its unreliable geometry.

To address this issue, we propose a SMEL framework, as illustrated in Fig. 2. Instead of further increasing encoder complexity or stacking additional constraints around a single evidence source, the proposed method first organizes complementary evidence from spectral characteristics, region context, and region prototype relations before assignment learning. These evidence sources are then processed by a lightweight multi-evidence assignment learning module, which preserves evidence-specific geometry through branch-private low-rank encoding and compares different evidence sources in a shared assignment space. In this way, different spectral–spatial cues can contribute to a common clustering decision without being prematurely mixed in the feature space.

B. Structured Multi-Evidence Organization

The structured multi-evidence organization follows three principles: compactness, complementarity, and alignability. First, each evidence source should provide compact information rather than high-dimensional redundant descriptors. Second, different evidence sources should describe complementary spectral–spatial properties of the same HSI, instead of repeatedly encoding the same cue. Third, the constructed evidence representations should remain sufficiently related, so that they can be compared in a shared assignment space. Following these principles, we instantiate three representative evidence sources for HSI clustering, corresponding to spectral shape, region context, and region prototype relations.

Let the HSI contain N pixels, and denote the raw spectrum of the ith pixel as $\mathbf{x}_i \in \mathbb{R}^B$, where B is the number of spectral bands. Following the above principles, we construct $V = 3$ structured evidence representations from the same HSI, denoted as $\{\mathbf{X}^{(v)}\}_{v=1}^{V}$, where $\mathbf{X}^{(v)} \in \mathbb{R}^{N\times d_v}$ and d_v denotes the feature dimension of the vth evidence source. In addition, we obtain a superpixel partition $\{S_m\}_{m=1}^{M}$ using entropy rate superpixel (ERS) segmentation [42], where $M \ll N$. The ERS partition provides local region units for constructing regional context, regional prototypes, and region-aware training targets. The constructed evidence representations are not independent modalities, but different organizations of the same spectral–spatial data. Such a design keeps the evidence sources related to the same clustering task while allowing each source to capture evidence-specific contextual information.

1) Spectral Evidence: The first evidence source focuses on reducing spectral redundancy while preserving the overall spectral profile. Since adjacent HSI bands are usually highly correlated, we partition the original B bands into $g = 16$ groups and perform groupwise average pooling. This operation suppresses local band-level redundancy and noise while retaining the coarse spectral shape. In addition, we compute the first-order spectral difference to describe local variations between adjacent bands.

Let $\mathcal{P}_g(\cdot)$ denote groupwise spectral pooling with g groups, and let $\Delta\mathbf{x}_i$ denote the first-order spectral difference of $\mathbf{x}_i$. The spectral evidence is defined as

$$\mathbf{x}_i^{(1)} = [\mathcal{P}_{16}(\mathbf{x}_i), \mathcal{P}_{16}(\Delta\mathbf{x}_i)] \in \mathbb{R}^{d_1}. \tag{1}$$

This spectral evidence provides a compact description that is closest to the original spectral signal and is therefore used as a compact spectral reference for later contextual evidence construction.

2) Region Context Evidence: The second evidence source introduces region context into a compact spectral representation. We first use coarser groupwise pooling with $g = 8$ to retain the global spectral shape at a lower dimension. To compensate for possible information loss caused by pooling, we also include a PCA representation of the raw spectrum, denoted as $\mathrm{PCA}_{20}(\mathbf{x}_i) \in \mathbb{R}^{20}$, which captures global linear subspace information.

In addition, we introduce superpixel-level regional statistics. Let $s(i) \in \{1, \ldots, M\}$ denote the superpixel index of pixel i, where M is the number of superpixels. Define a compact spectral representation $\mathbf{h}_i = [\mathcal{P}_8(\mathbf{x}_i), \mathcal{P}_8(\Delta\mathbf{x}_i)]$. The corresponding superpixel-level region prototype is

$$\bar{\mathbf{h}}_{s(i)} = \frac{1}{|S_{s(i)}|}\sum_{j\in S_{s(i)}} \mathbf{h}_j \tag{2}$$

where $S_m = \{i : s(i) = m\}$ is the set of pixels in the mth superpixel. The region context evidence is then constructed as

$$\mathbf{x}_i^{(2)} = \left[\mathrm{PCA}_{20}(\mathbf{x}_i), \mathcal{P}_8(\mathbf{x}_i), \mathcal{P}_8(\Delta\mathbf{x}_i), \bar{\mathbf{h}}_{s(i)}\right] \in \mathbb{R}^{d_2}. \tag{3}$$

This evidence source combines global PCA information, coarsely grouped spectral characteristics, spectral variations, and regional statistics to provide a heterogeneous global–local contextual description.

3) Region Prototype Evidence: The third evidence source uses the superpixel-level prototype as a local regional reference. Let $\mathbf{b}_i = [\mathcal{P}_{16}(\mathbf{x}_i), \mathcal{P}_{16}(\Delta\mathbf{x}_i)] \in \mathbb{R}^{d_b}$ be the compact spectral basis. The corresponding region prototype is $\bar{\mathbf{b}}_{s(i)} = 1/|S_{s(i)}| \sum_{j\in S_{s(i)}} \mathbf{b}_j$. The region prototype evidence is defined as

$$\mathbf{x}_i^{(3)} = \left[\mathbf{b}_i, \bar{\mathbf{b}}_{s(i)}\right] \in \mathbb{R}^{d_3}. \tag{4}$$

Different from the heterogeneous contextual organization above, the pixel descriptor and its regional prototype are represented in the same finer-grained spectral coordinate system,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

providing a matched regional reference while retaining pixel-level spectral details.

In summary, the three evidence sources differ not simply in whether they use spectral or spatial information but in how they organize spectral–spatial context. The spectral evidence provides compact spectral characteristics, the region context evidence injects regional statistics into heterogeneous spectral descriptions, and the region prototype evidence uses superpixel prototypes as local references for pixel-level descriptors. This design reduces the influence of one unreliable evidence source while preventing the constructed evidence sources from becoming unrelated feature spaces that are difficult to compare in a shared assignment space.

C. Lightweight Multi-Evidence Assignment Learning

After constructing complementary evidence sources, the next challenge is how to learn clustering assignments from them in a comparable and lightweight manner. Although these evidence sources are derived from the same HSI, they are not statistically equivalent. Direct concatenation may mix heterogeneous descriptors, whereas independently parameterized heads introduce redundant output mappings and do not provide an explicitly shared parameterization across evidence branches.

To address this issue, we develop a lightweight multi-evidence assignment learning mechanism with branch-private low-rank encoding and shared assignment projection. The branch-private encoder preserves evidence-specific geometry in compact subspaces, while the shared assignment projection maps different evidence-specific embeddings into a common clustering coordinate system. This design avoids both early feature-level overfusion and late independent clustering.

Specifically, each evidence source is first mapped into its own low-rank subspace. The branch-private low-rank encoder is introduced not merely to reduce parameters but to provide a stable evidence-specific bottleneck. Since contiguous spectral bands are highly correlated and regional evidence is induced by limited land-cover structures within a single scene, projecting each evidence source into a compact low-rank subspace can capture its dominant structure while avoiding direct high-dimensional feature mixing.

For the vth evidence source $\mathbf{X}^{(v)}$, given the current low-rank basis $\mathbf{A}^{(v)}$, we consider the following ridge-coding problem:

$$\mathbf{Z}_0^{(v)}\left(\mathbf{A}^{(v)}\right) = \arg\min_{\mathbf{Z}\in\mathbb{R}^{N\times r}} \left\|\mathbf{X}^{(v)} - \mathbf{Z}\left(\mathbf{A}^{(v)}\right)^{\top}\right\|_F^2 + \beta\|\mathbf{Z}\|_F^2 \tag{5}$$

where $\mathbf{A}^{(v)} \in \mathbb{R}^{d_v\times r}$ is the low-rank basis matrix, $\mathbf{Z}_0^{(v)} \in \mathbb{R}^{N\times r}$ is the corresponding base code matrix, r is the rank dimension, and $\beta > 0$ is the ridge regularization coefficient. We initialize $\mathbf{A}^{(v)}$ using the leading r directions obtained from truncated SVD of a centered sampled version of $\mathbf{X}^{(v)}$ and then treat it as a learnable parameter during training.

Given $\mathbf{A}^{(v)}$, the closed-form ridge solution is

$$\mathbf{Z}_0^{(v)} = \mathbf{X}^{(v)}\mathbf{A}^{(v)}\left[\left(\mathbf{A}^{(v)}\right)^{\top}\mathbf{A}^{(v)} + \beta\mathbf{I}_r\right]^{-1} \in \mathbb{R}^{N\times r} \tag{6}$$

where $\mathbf{I}_r \in \mathbb{R}^{r\times r}$ is the identity matrix. For a fixed $\mathbf{A}^{(v)}$, the ridge objective is strictly convex with respect to $\mathbf{Z}$, and thus, (6) gives its unique solution. During each forward pass, $\mathbf{Z}_0^{(v)}$ is recomputed from the current basis rather than treated as an independent learnable variable, and the downstream clustering losses are backpropagated through this closed-form mapping to update $\mathbf{A}^{(v)}$.

The ith row of $\mathbf{Z}_0^{(v)}$ is denoted as $\mathbf{z}_{i,0}^{(v)}$, which is further refined by a lightweight residual adapter

$$z_i^{(v)} = \mathrm{LN}\left(z_{i,0}^{(v)} + \gamma f_\theta^{(v)}\left(z_{i,0}^{(v)}\right)\right) \tag{7}$$

where $f_\theta^{(v)}(\cdot)$ is a two-layer MLP, γ is the residual scaling factor, and $\mathrm{LN}(\cdot)$ denotes layer normalization. The refined embeddings of all pixels are collected as $\mathbf{Z}^{(v)} = [\mathbf{z}_1^{(v)}, \ldots, \mathbf{z}_N^{(v)}]^{\top}$. The closed-form ridge code gives a stable base representation that follows the dominant linear structure of each evidence source, reducing the instability caused by fully random deep encoders in the unsupervised setting. The residual adapter then performs limited task-adaptive refinement in the low-rank space, allowing the model to adjust the base evidence geometry without completely overwriting it. In implementation, the last linear layer of the residual adapter is initialized to zero, so that each branch starts close to the ridge low-rank encoding.

After low-rank encoding, a shared assignment projection is applied to all evidence branches, so that their predictions are expressed in a common clustering coordinate. This parameter-sharing design enables direct assignment-level integration while preserving the evidence-specific information in the branch-private embeddings.

Let K be the number of clusters. For each evidence source, the assignment probability of pixel i is computed as

$$\mathbf{p}_i^{(v)} = \mathrm{softmax}\left(\mathbf{W}_c\mathbf{z}_i^{(v)} + \mathbf{b}_c\right) \in \mathbb{R}^K \tag{8}$$

where $\mathbf{W}_c \in \mathbb{R}^{K\times r}$ and $\mathbf{b}_c \in \mathbb{R}^K$ are shared across all evidence sources. Let

$$\mathbf{P}^{(v)} = \left[\mathbf{p}_1^{(v)}, \ldots, \mathbf{p}_N^{(v)}\right]^{\top} \in \mathbb{R}^{N\times K} \tag{9}$$

denote the assignment matrix of the vth evidence source. The fused assignment probability is obtained by averaging evidence-specific predictions

$$\mathbf{P} = \frac{1}{V}\sum_{v=1}^{V}\mathbf{P}^{(v)} \in \mathbb{R}^{N\times K}. \tag{10}$$

Therefore, the shared assignment projection serves as a common clustering coordinate rather than an independent classifier for each evidence branch. It does not require the evidence-specific predictions to be identical: the shared head provides common output parameterization, while the warm-up teacher promotes semantic alignment across evidence branches.

D. Training Strategy

Although the shared assignment projection provides a common clustering coordinate, the semantic correspondence of this coordinate still needs to be established without ground-truth labels. Directly training a randomly initialized assignment head may lead to unstable assignments, especially when multiple evidence sources provide different clustering cues. We therefore adopt a two-stage training strategy. The first stage

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

uses a superpixel-refined teacher to establish a stable initial assignment coordinate shared by all evidence branches. The second stage refines pixel-level assignments using soft anchor guidance and confidence-aware regional targets.

1) Stage-I: Warm-Up: We first generate a superpixel-refined teacher partition to establish an initial assignment coordinate. Using the ERS superpixel partition $\{S_m\}_{m=1}^{M}$ defined above, we build a teacher feature by concatenating the standardized evidence representations. A region prototype is then computed for each superpixel in this teacher feature space. We perform size-weighted K-means on these region prototypes, where the weight of each prototype is proportional to the number of pixels in the corresponding region. The obtained centers are used to initialize pixel-level K-means, producing a hard teacher label vector $\hat{\mathbf{y}} = \{\hat{y}_i\}_{i=1}^{N}$, $\hat{y}_i \in \{1, \ldots, K\}$. Compared with directly applying randomly initialized K-means to all pixels, this strategy first searches for stable cluster centers at the superpixel-prototype level and then refines pixel-level assignments in the teacher feature space. It therefore reduces the influence of pixel-level noise and mixed pixels while preserving pixel-level clustering resolution.

During warm-up, each evidence branch is trained with the same teacher partition to establish a common assignment coordinate

$$\mathcal{L}_{\text{warm}} = \frac{1}{V}\sum_{v=1}^{V} \text{CE}\left(\hat{\mathbf{y}}, \mathbf{P}^{(v)}\right) \tag{11}$$

where $\text{CE}(\hat{\mathbf{y}}, \mathbf{P}^{(v)})$ denotes the cross-entropy between the hard teacher labels and the evidence-specific prediction.

After warm-up, the fused assignment probability computed by (10) is fixed as the soft anchor reference $\mathbf{P}_0 \in \mathbb{R}^{N\times K}$. Different from the hard teacher label vector $\hat{\mathbf{y}}$, $\mathbf{P}_0$ is a soft prediction produced by the warmed-up multi-evidence assignment model. It retains sample-level uncertainty and cross-evidence confidence information and is therefore used as a stable reference for the subsequent refinement stage.

2) Stage-II: Anchor-Region Refinement: After warm-up, the model obtains a fixed soft anchor reference $\mathbf{P}_0 = [\mathbf{p}_1^0, \ldots, \mathbf{p}_N^0]^\top \in \mathbb{R}^{N\times K}$. Directly optimizing the model with its online predictions may cause assignment drift, while strictly following $\mathbf{P}_0$ may preserve possible errors from the initial teacher. Therefore, Stage-II uses two complementary objectives: a confidence-weighted anchor loss to maintain a stable assignment coordinate and a regional mixed-target loss to refine local predictions with superpixel-level consistency while allowing uncertain pixels to deviate from unreliable regional cues.

Let $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_N]^\top$ be the current fused assignment probability computed by (10). The anchor loss aligns the current prediction with the soft anchor reference

$$\mathcal{L}_{\text{anchor}} = -\frac{\sum_{i=1}^{N} w_i \sum_{k=1}^{K} p_{ik}^0 \log p_{ik}}{\sum_{i=1}^{N} w_i} \tag{12}$$

where p_{ik}^0 and p_{ik} are the kth elements of $\mathbf{p}_i^0$ and $\mathbf{p}_i$, respectively. The confidence weight is defined as

$$w_i = \text{clip}\left(\frac{\max_k p_{ik}^0 - c_{\text{low}}^a}{c_{\text{high}}^a - c_{\text{low}}^a}, 0, 1\right) \tag{13}$$

where c_{low}^a and c_{high}^a are set to 0 and 0.8, respectively. This weighting strategy weakens unreliable anchor assignments while preserving high-confidence assignment coordinates.

To further exploit local spatial consistency, we construct a regional mixed target for each superpixel. For the mth superpixel S_m, the anchor-based regional distribution and the current regional consensus are computed as

$$\mathbf{q}_m^0 = \mathcal{T}_\tau\left(\frac{1}{|S_m|}\sum_{i\in S_m}\mathbf{p}_i^0\right), \quad \mathbf{q}_m^c = \mathcal{T}_\tau\left(\frac{1}{|S_m|}\sum_{i\in S_m}\mathbf{p}_i\right)$$

where $\mathcal{T}_\tau(\cdot)$ denotes the temperature-sharpening operator defined by

$$\left[\mathcal{T}_\tau(\mathbf{p})\right]_k = \frac{(\max(p_k, \epsilon))^{1/\tau}}{\sum_{j=1}^{K}\left(\max\left(p_j, \epsilon\right)\right)^{1/\tau}} \tag{14}$$

where $\tau = 0.5$ is the temperature and $\epsilon = 10^{-8}$ is used for numerical stability. Since $\tau < 1$, this operation sharpens the input probability distribution. When $\mathbf{q}_m^c$ is used as a target, the current predictions are detached to avoid trivial self-reinforcement. The confidence of the anchor-based regional distribution is measured by

$$\alpha_m = \text{clip}\left(\frac{\max_k q_{mk}^0 - c_{\text{low}}^r}{c_{\text{high}}^r - c_{\text{low}}^r}, 0, 1\right) \tag{15}$$

where $c_{\text{low}}^r = 0.55$ and $c_{\text{high}}^r = 0.9$. The final regional target is

$$\mathbf{q}_m = \alpha_m \mathbf{q}_m^0 + (1-\alpha_m)\,\mathbf{q}_m^c. \tag{16}$$

The regional mixed-target loss is then defined as

$$\mathcal{L}_{\text{reg}} = \frac{1}{VN}\sum_{v=1}^{V}\sum_{i=1}^{N}\text{CE}\left(\mathbf{q}_{s(i)}, \mathbf{p}_i^{(v)}\right) \tag{17}$$

where $s(i)$ denotes the superpixel index of pixel i. Unlike hard superpixel smoothing, this loss does not force all pixels in a superpixel to share the same label. Instead, it uses a soft regional target that combines the stable anchor reference and the current regional consensus, thereby improving local consistency while retaining uncertainty around boundaries and mixed pixels. The final objective of Stage-II is

$$\mathcal{L}_{\text{main}} = \lambda_a \mathcal{L}_{\text{anchor}} + \lambda_r \mathcal{L}_{\text{reg}} \tag{18}$$

where λ_a and λ_r are the weights of the anchor loss and the regional mixed-target loss, respectively.

To remove isolated predictions, we apply a weighted four-neighborhood readout. The normalized spectrum and the weight between adjacent pixels are defined as

$$\tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i}{\max\left(\|\mathbf{x}_i\|_2, \epsilon\right)}, \quad w_{ij} = \exp\left(-\frac{\left\|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\right\|_2^2}{\sigma_{\text{read}}}\right), j \in \mathcal{N}_4(i).$$

Starting from $l_i^{(0)} = \arg\max_c P_{i,c}$, the labels are synchronously updated by

$$E_i^{(t)}(c) = -\log\max\left(P_{i,c}, \epsilon\right) + \lambda_{\text{read}}\sum_{j\in\mathcal{N}_4(i)} w_{ij}\mathbb{I}\left(l_j^{(t)} \neq c\right).$$

At each iteration, all labels are synchronously updated as $l_i^{(t+1)} = \arg\min_c E_i^{(t)}(c)$. Here, $\mathcal{N}_4(i)$ denotes the four-connected neighborhood. We set $\epsilon = 10^{-8}$, $\sigma_{\text{read}} = 0.08$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Structured Multi-Evidence Learning

Input: Hyperspectral image $\mathbf{X}$, number of clusters K, superpixel partition $\{S_m\}_{m=1}^{M}$, rank r.

Output: Final clustering map $\tilde{\mathbf{y}}$.

Model construction:

Construct structured evidence representations $\{\mathbf{X}^{(v)}\}_{v=1}^{V}$ through (a) spectral evidence, (b) region context evidence, and (c) region prototype evidence.

Obtain evidence-specific embeddings $\{\mathbf{Z}^{(v)}\}_{v=1}^{V}$ using the lightweight low-rank encoder in Eqs. (6)–(7).

Produce evidence-specific assignments $\{\mathbf{P}^{(v)}\}_{v=1}^{V}$ with the shared assignment head, and fuse by Eq. (10).

Two-stage training:

Generate the superpixel-refined teacher labels $\hat{\mathbf{y}}$, and warm up all evidence branches with Eq. (11).

Fix the warmed-up fused prediction as $\mathbf{P}_0$, and refine the model with Eq. (18).

Inference:

Compute the final fused assignment probability $\mathbf{P}$, and obtain $\tilde{\mathbf{y}}$ by the weighted spatial readout.

return $\tilde{\mathbf{y}}$.

and $\lambda_{\text{read}} = 35$ for all datasets. The readout terminates when the labels remain unchanged or after ten iterations and is applied only at inference without updating the model parameters.

E. Parameter and Computational Complexity

The lightweight design mainly comes from the low-rank encoder and the shared assignment head. For the vth evidence source with dimension d_v, the learnable basis matrix $\mathbf{A}^{(v)} \in \mathbb{R}^{d_v \times r}$ contains $d_v r$ parameters, and the two-layer residual adapter in the r-dimensional space contains approximately $2r^2 + O(r)$ parameters. Since the assignment head is shared by all evidence sources, it introduces only $rK + K$ parameters. Therefore, the total number of learnable parameters is

$$N_{\text{param}} = \sum_{v=1}^{V} d_v r + V\left(2r^2 + O(r)\right) + rK + K.$$

This number is independent of the number of pixels N. Since r is small and each constructed evidence source has a compact feature dimension, the proposed model remains lightweight even with multiple evidence sources.

For a mini-batch with B_s samples, the main computational cost is

$$O\left(B_s \sum_{v=1}^{V} d_v r + B_s V r^2 + B_s V r K + \sum_{v=1}^{V} d_v r^2 + V r^3\right).$$

The first three terms correspond to low-rank encoding, residual adaptation, and assignment prediction, respectively. The last two terms come from forming the low-rank projection matrices and inverting the small $r \times r$ ridge matrices. Since r is small in practice, the proposed model maintains a low parameter scale and moderate computational cost.

TABLE I
SUMMARY OF THE BENCHMARK HYPERSPECTRAL DATASETS

Dataset	Spatial Size	Bands	Classes
Pavia University (PU)	610 × 340	103	9
Botswana (BO)	1476 × 256	145	14
Trento (TR)	600 × 166	63	6
Pavia Center (PC)	1096 × 715	102	9
LongKou (LK)	550 × 400	270	9

IV. EXPERIMENTS

A. Experimental Settings

To evaluate the effectiveness of the proposed method under different hyperspectral scenes, we conduct experiments on five widely used HSI datasets, including Pavia University (PU), Botswana (BO), Trento (TR), Pavia Center (PC), and LongKou (LK). These datasets cover urban, agricultural, and natural scenes with different spatial sizes, spectral bands, and class numbers, providing a comprehensive benchmark for unsupervised hyperspectral clustering. The basic information of the datasets is summarized in Table I.

We compare the proposed method with several representative unsupervised hyperspectral clustering methods, including NCSC [33], DSCRLE [43], S2GCL [44], SSGCC [45], SDST [13], BGPC [25], SLCGC [40], SAPC [9], LFGD [46], EAGLE [12], and SSSC [47]. These methods cover different technical routes, such as graph learning, contrastive learning, subspace modeling, and deep clustering, and thus provide a broad comparison for evaluating clustering accuracy, stability, and model complexity. Since the official implementation of the recent S^2ESCC [41] model is not publicly available, we include Sub-PCA20 only as an evidence proxy to examine the effect of spectral subview partitioning. To reduce the influence of random initialization, we report the average results over five independent runs. Three commonly used clustering metrics are adopted, including clustering accuracy (ACC), normalized mutual information (NMI), and adjusted Rand index (ARI). All these metrics range from 0 to 1, and larger values indicate better agreement between the predicted clustering labels and the ground-truth categories. Specifically, ACC is computed after optimal label matching, NMI measures the shared information between the clustering result and the ground truth, and ARI evaluates the pairwise consistency between two partitions while correcting for random agreement.

To ensure a fair comparison, we follow the official implementations and parameter settings of the compared methods whenever available. For methods whose public code or dataset-specific settings are incomplete, we tune the parameters within the recommended ranges in the original papers and report the best reproducible results. All experiments are performed on a Windows 11 laptop equipped with an AMD Ryzen 7 7840HS CPU and 64 GB of memory. The implementation is available at https://github.com/NuclearLemon/SMEL. Unless otherwise specified, our method uses the same training configuration for all datasets: 50 warm-up epochs, 100 main epochs, mini-batch size 1024, Adam optimizer with learning rate 10^{-3} and weight decay 10^{-4}, and gradient clipping threshold 5. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

TABLE II
DIAGNOSIS OF DIFFERENT STRUCTURAL EVIDENCE SOURCES AND CONSTRUCTED EVIDENCE REPRESENTATIONS IN TERMS OF ACC

Dataset	Raw	PCA20	Sub-PCA20	Spectral mean	Spectral slope	SP mean	SP graph	SP graph weighted	Spectral evidence	Region context evidence	Region prototype evidence	Naive concat
PU	0.565	0.552	0.534	0.554	0.568	0.391	0.470	0.477	0.596	0.630	0.520	0.658
BO	0.575	0.512	0.531	0.541	0.450	0.477	0.593	0.638	0.624	0.647	0.685	0.659
TR	0.645	0.612	0.610	0.650	0.687	0.875	0.852	0.866	0.615	0.778	0.823	0.795
PC	0.749	0.860	0.881	0.749	0.780	0.707	0.866	0.807	0.765	0.870	0.886	0.796
LK	0.699	0.682	0.739	0.697	0.707	0.684	0.639	0.673	0.739	0.751	0.737	0.772
Mean	0.647	0.643	0.659	0.638	0.638	0.627	0.684	0.692	0.668	0.735	0.730	0.736

PCA dimension used for evidence construction is fixed to 20. The ridge coefficient and residual scaling factor are fixed to $\beta = 10^{-2}$ and $\gamma = 0.2$, respectively. Unless otherwise stated, the two main-stage loss weights are fixed to $\lambda_a = 0.2$ and $\lambda_r = 0.5$ for all datasets. These common moderate values were selected once from a coarse preliminary range and were not tuned separately for individual datasets. For dataset-dependent structural settings, only the number of superpixels and the representation rank are adjusted according to the spatial scale and class complexity of each dataset. The detailed values are provided in the released code.

B. Diagnosis of Structural Evidence Reliability

We first conduct a diagnostic experiment to examine the reliability of different structural evidence sources. This experiment is not intended to evaluate the final model. Instead, it aims to analyze a more fundamental question: whether individual spectral, regional, or graph-based evidence sources can provide stable clustering evidence for hyperspectral data. Specifically, we examine the following structural evidence sources and constructed evidence representations.

1) *Raw Spectra:* Standardized original spectra.
2) *PCA20:* The first 20 principal components.
3) *Sub-PCA20:* PCA features extracted from two contiguous spectral subviews and concatenated into a 20-D representation.
4) *Spectral Mean:* Groupwise averages over consecutive bands.
5) *Spectral Slope:* Groupwise averages of first-order spectral differences.
6) *SP Mean:* Superpixel-averaged spectra.
7) *SP Graph:* Uniform aggregation over a k-nearest-neighbor superpixel graph.
8) *SP Graph Weighted:* Similarity-weighted aggregation over the superpixel graph.
9) *Spectral Evidence:* The proposed combination of grouped spectral means and slopes.
10) *Region Context Evidence:* The proposed combination of PCA features, coarse spectral pooling, spectral slopes, and regional context.
11) *Region Prototype Evidence:* The proposed combination of pixel descriptors and their superpixel prototypes.
12) *Naive Concat:* Direct concatenation of the three proposed evidence representations.

The descriptor-level diagnostic experiment uses a common SLIC partition with an automatic target region area of 200 pixels for all datasets. The concatenated evidence is standardized and evaluated by K-means.

Table II shows that different structural evidence sources exhibit clear scene-dependent behavior. Raw spectra and PCA20 preserve global spectral information, but they rely mainly on spectral similarity and therefore may be insufficient under spectral mixing, large intraclass variation, or weak interclass separability. Sub-PCA20 slightly improves the average performance over PCA20 by introducing spectral subview information, but the gain is still dataset dependent. Similarly, the spectral mean and spectral slope capture coarse spectral shape and local spectral variation, respectively, yet neither of them consistently improves clustering across all scenes. These results indicate that individual spectral evidence alone cannot provide universally reliable clustering evidence.

The spatial and graph-based evidence sources also show strong dependence on the reliability of the underlying regional structure. SP mean can suppress pixel-level noise when superpixels are semantically coherent, but it may weaken discriminative information around boundaries, small objects, and mixed regions. The SP graph and SP graph weighted further introduce regional neighborhood propagation, which is beneficial when graph edges reflect semantic relations. However, when superpixel partitions or graph neighborhoods are inaccurate, graph smoothing may propagate unreliable information and degrade clustering quality. This explains why graph-based evidence sources perform well on some datasets but remain unstable on others.

Compared with individual structural evidence sources, the constructed evidence representations provide more reliable clustering cues on average, especially region context evidence and region prototype evidence. Specifically, region context evidence performs better on PU and LK, whereas region prototype evidence achieves higher ACC on BO, TR, and PC, indicating that the two representations provide nonidentical, scene-dependent regional descriptions. This suggests that spectral shape, regional context, and region prototype relations provide complementary information for hyperspectral clustering. Naive concat further confirms the usefulness of combining complementary evidence, achieving the highest mean ACC among the fixed evidence settings. Nevertheless, it is not consistently the best choice on each dataset, and it still performs K-means in a fixed concatenated feature space. These observations suggest that combining complementary evidence

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

TABLE III
CLUSTERING RESULTS ON FIVE BENCHMARK HYPERSPECTRAL DATASETS

Dataset	Metric	NCSC TGRS'22 [33]	DSCRLE TGRS'23 [43]	S2GCL TGRS'24 [44]	SSGCC TGRS'24 [45]	SDST TGRS'24 [13]	BGPC TMM'24 [25]	SLCGC TMM'25 [40]	SAPC TCSVT'25 [9]	LFGD KBS'25 [46]	EAGLE TIP'25 [12]	SSSC ICASSP'26 [47]	Ours
PU	ACC	0.550	0.514	0.648	0.515	0.570	**0.710**	0.663	0.639	0.663	0.662	0.428	0.709
	NMI	0.409	0.505	0.569	0.431	0.487	0.649	0.619	0.619	0.601	0.590	0.452	**0.717**
	ARI	0.410	0.302	0.553	0.343	0.359	**0.570**	0.502	0.449	0.476	0.488	0.243	**0.570**
BO	ACC	0.499	0.466	0.523	0.578	0.607	0.569	0.623	0.635	0.567	0.736	0.495	**0.776**
	NMI	0.628	0.565	0.641	0.647	0.694	0.716	0.734	0.709	0.692	0.805	0.642	**0.856**
	ARI	0.404	0.335	0.372	0.454	0.490	0.473	0.520	0.486	0.467	0.641	0.384	**0.719**
TR	ACC	0.801	0.839	0.803	0.889	0.799	0.838	0.817	0.738	0.805	0.901	0.905	**0.930**
	NMI	0.718	0.727	0.752	0.839	0.683	0.819	0.858	0.765	0.790	0.860	0.832	**0.868**
	ARI	0.757	0.739	0.751	0.914	0.641	0.828	0.824	0.680	0.773	0.890	0.893	**0.923**
PC	ACC	0.752	0.739	0.850	0.856	0.846	0.911	0.921	0.933	0.793	0.746	0.779	**0.943**
	NMI	0.592	0.717	0.815	0.738	0.791	0.870	0.877	0.910	0.719	0.614	0.635	**0.918**
	ARI	0.759	0.788	0.878	0.885	0.918	0.936	0.961	**0.974**	0.841	0.729	0.799	0.971
LK	ACC	0.513	0.559	0.688	0.756	0.810	0.717	0.677	0.829	0.696	0.736	0.594	**0.870**
	NMI	0.462	0.602	0.740	0.685	0.759	**0.827**	0.678	0.800	0.681	0.629	0.670	**0.827**
	ARI	0.327	0.490	0.671	0.727	0.802	0.686	0.667	0.755	0.548	0.653	0.533	**0.841**

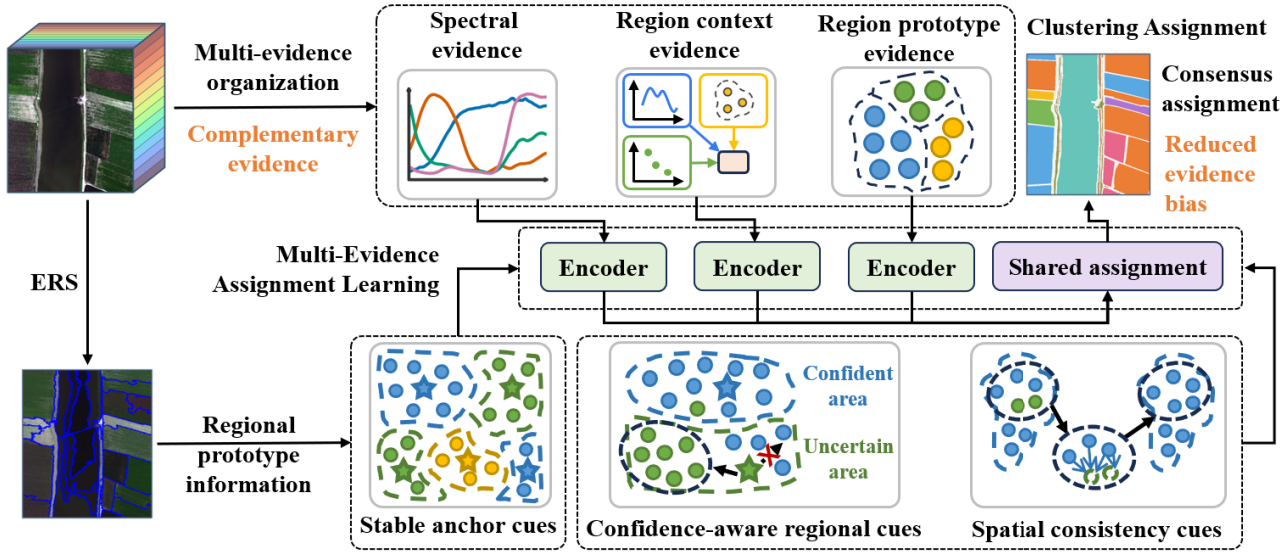


Fig. 2. Overall framework of the proposed SMEL method. Complementary evidence sources are constructed from a single HSI and mapped into a unified assignment space through lightweight low-rank branches and a shared assignment head, enabling assignment-level consensus while preserving evidence-specific information.

is useful, but fixed feature-space concatenation is not sufficient to fully organize complementary evidence sources.

C. Main Results

To evaluate the overall effectiveness of the proposed method, we compare it with representative unsupervised hyperspectral clustering methods. Table III reports the ACC, NMI, and ARI results on five benchmark datasets. The best and second-best results are highlighted in bold and underlined, respectively. The proposed method achieves the best performance on 13 out of 15 metrics and obtains the second-best result on the remaining two metrics. In particular, it consistently improves the clustering accuracy on Botswana, Trento, PC, and LongKou and achieves nearly the best ACC on PU with only a marginal gap of 0.001 compared with BGPC. These results show that the proposed method provides strong and stable clustering performance across different hyperspectral scenes.

Due to space limitations, we provide representative visual comparisons on the PC and LongKou datasets in Figs. 3 and 4, respectively. Compared with several competing methods, the proposed method produces a clustering map that is more spatially coherent and closer to the ground-truth land-cover distribution. Many compared methods can recover the rough spatial layout, but they still suffer from fragmented regions, noisy local assignments, or confusion between spectrally similar categories. In contrast, our method better preserves homogeneous regions while reducing isolated noisy predictions. This visual comparison is consistent with the quantitative results in Table III, where the proposed method achieves the best ACC, NMI, and ARI on LongKou.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

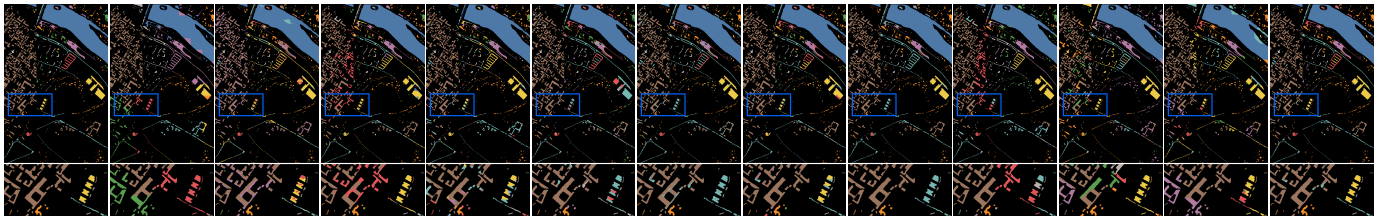


Fig. 3. Visual comparison of clustering maps on the PC dataset. From left to right: (a) GT, (b) NCSC, (c) DSCRLE, (d) S2GCL, (e) SSGCC, (f) SDST, (g) BGPC, (h) SLCGC, (i) SAPC, (j) LFGD, (k) EAGLE, (l) SSSC, and (m) ours.

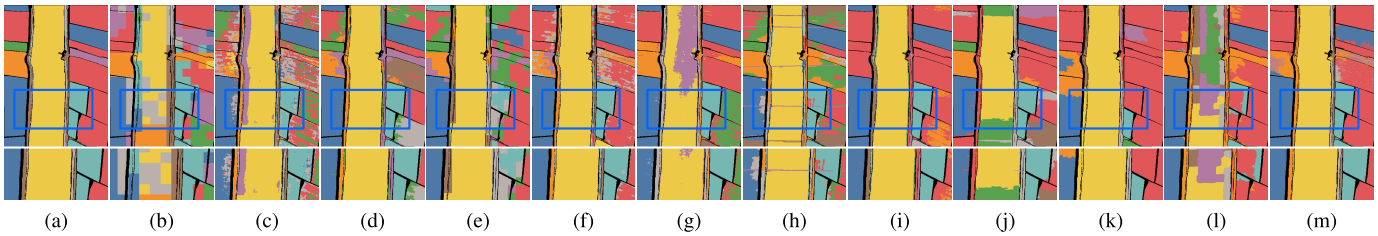


Fig. 4. Visual comparison of clustering maps on the LongKou dataset. From left to right: (a) GT, (b) NCSC, (c) DSCRLE, (d) S2GCL, (e) SSGCC, (f) SDST, (g) BGPC, (h) SLCGC, (i) SAPC, (j) LFGD, (k) EAGLE, (l) SSSC, and (m) ours.

TABLE IV
COMPONENT ABLATION RESULTS OF THE PROPOSED METHOD

Dataset	*Concat-KM*			*SMEL-PixelT*			*FeatureFuse*			*SMEL-Stage-I*			*SMEL-IndHead*			*SMEL*		
	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
PU	0.645	0.638	0.477	0.693	0.708	0.551	0.681	0.698	0.523	0.703	0.713	0.552	0.704	0.716	0.556	0.709	0.717	0.570
BO	0.647	0.758	0.567	0.684	0.821	0.631	0.731	0.817	0.659	0.763	0.827	0.686	0.773	0.855	0.717	0.776	0.856	0.719
TR	0.783	0.703	0.655	0.930	0.869	0.924	0.903	0.823	0.872	0.869	0.789	0.793	0.930	0.870	0.924	0.930	0.868	0.923
PC	0.800	0.836	0.873	0.904	0.889	0.962	0.937	0.893	0.957	0.937	0.907	0.963	0.943	0.918	0.971	0.943	0.918	0.971
LK	0.795	0.779	0.789	0.907	0.863	0.896	0.845	0.823	0.815	0.830	0.811	0.814	0.872	0.832	0.847	0.870	0.827	0.841

Compared with the existing methods, the advantage of our method is not limited to a specific dataset or metric. Several compared methods achieve competitive results on certain scenes. However, their performance is less balanced across all datasets. This is because many existing methods construct self-supervised signals around a specific input evidence, graph structure, contrastive relation, propagation rule, or learned embedding. When the underlying evidence or structural relation is less reliable for a particular scene, the learned clustering representation may also become unstable. In contrast, the proposed method constructs complementary structural evidence from spectral shape, region context, and region prototype relations and further organizes them in a unified assignment space. This design helps reduce the influence of unreliable single evidence and leads to more consistent performance.

The main results are also consistent with the evidence reliability diagnosis in Table II. Individual evidence sources, such as raw spectra, regional averages, and superpixel graph smoothing, cannot stably dominate across all datasets, while naive concatenation only combines multiple evidence sources in a fixed feature space. By comparison, the proposed framework improves clustering performance by jointly using multi-evidence construction and assignment-level organization. This indicates that the gain comes not only from constructing complementary evidence but also from mapping such evidence into a shared clustering space through the lightweight assignment learning framework.

D. Component Ablation

To analyze the contribution of each component, we conduct an ablation study on evidence fusion, teacher construction, training stages, and assignment-head parameterization. The results are reported in Table IV.

The six variants are defined as follows: 1) *Concat-KM* directly concatenates the three evidence representations and applies K-means without network training; 2) *SMEL-PixelT* replaces the superpixel-refined teacher with pixel-level K-means labels; 3) *FeatureFuse* concatenates the three branch embeddings before applying a single assignment head; 4) *SMEL-Stage-I* removes the second training stage; 5) *SMEL-IndHead* replaces the shared assignment head with three independent heads; and 6) *SMEL* denotes the complete model.

1) Evidence Fusion: Concat-KM directly concatenates the constructed evidence representations before K-means. However, simple concatenation only increases the amount of input information and does not explicitly organize different evidence sources in a common assignment space. Therefore, its performance is still lower than the learning-based stages. Moreover, *SMEL-IndHead* consistently outperforms *FeatureFuse*, showing that heterogeneous evidence is more effectively integrated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION RESULTS OF THE MAIN-STAGE TRAINING LOSSES

Dataset	$\mathcal{L}_{\text{anchor}}$	$\mathcal{L}_{\text{reg}}$	ACC	NMI	ARI
PU		✓	0.690	0.664	0.551
	✓		0.686	0.678	0.535
	✓	✓	**0.709**	**0.717**	**0.570**
BO		✓	0.718	0.790	0.632
	✓		0.740	0.796	0.650
	✓	✓	**0.776**	**0.856**	**0.719**
TR		✓	0.911	0.830	0.877
	✓		0.808	0.720	0.681
	✓	✓	**0.930**	**0.868**	**0.923**
PC		✓	0.942	0.908	0.969
	✓		0.927	0.890	0.954
	✓	✓	**0.943**	**0.918**	**0.971**
LK		✓	0.809	0.797	0.793
	✓		0.815	0.786	0.797
	✓	✓	**0.870**	**0.827**	**0.841**

after branch-specific assignment prediction than through early feature-level fusion.

2) Teacher Construction: Compared with *SMEL-PixelT*, the complete model achieves stronger overall performance. The two variants differ only in teacher construction: pixel-level K-means is more susceptible to local spectral ambiguity and isolated noise, whereas the superpixel-refined teacher exploits regional statistics to obtain a more reliable initial partition.

3) Stage-II Refinement: Stage-I further improves the results by training all evidence branches with a shared assignment head under the superpixel-refined teacher. Compared with *SMEL-Stage-I*, the complete *SMEL* model consistently improves all evaluation metrics, demonstrating the value of the second-stage refinement. The anchor loss preserves the global assignment coordinate obtained after warm-up, while the regional mixed-target loss refines local predictions with superpixel-level consistency.

4) Assignment-Head Parameterization: *SMEL-IndHead* and *SMEL* achieve highly similar clustering results, indicating that independent assignment heads do not provide a consistent advantage. The shared-head design therefore provides a more compact parameterization, using only one-third of the assignment-head parameters.

E. Loss Ablation

We further conduct a loss ablation study to analyze the two objectives used in the main training stage, namely, the anchor loss and the regional mixed-target loss. Since the role of the warm-up stage has been examined in the component ablation, this experiment focuses on whether the two main-stage losses are both necessary after the initial assignment coordinate has been established.

As shown in Table V, using both losses consistently achieves the best performance on all datasets. When the anchor loss is removed, the model relies mainly on regional mixed targets. Although local spatial consistency can still provide useful guidance, the model lacks a stable global assignment reference and may overemphasize local smoothness, especially when superpixels contain mixed or boundary regions. This leads to performance degradation on all datasets.

When the regional mixed-target loss is removed, the model is guided only by the soft anchor reference. This helps preserve the global assignment coordinate but limits the ability to correct local unstable predictions. The performance drop is particularly evident on Trento and LongKou, indicating that regional consistency is important for suppressing local noise and improving spatially coherent clustering.

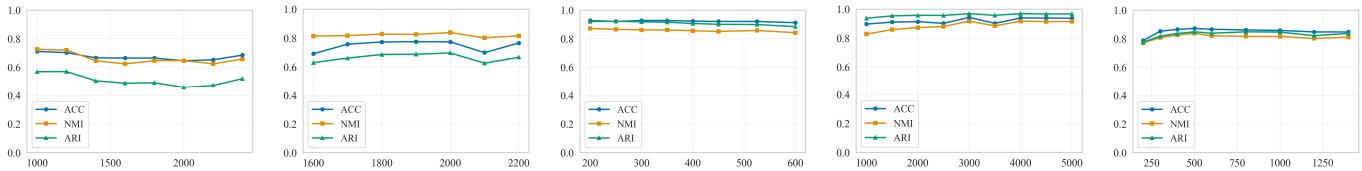
These results show that the two losses play complementary roles. The anchor loss stabilizes the global clustering coordinate inherited from warm-up, while the regional mixed-target loss refines local predictions using confidence-aware superpixel information. Their combination enables the model to maintain global assignment consistency and improve local evidence reliability, leading to more robust clustering results across different hyperspectral scenes.

F. Parameter Sensitivity Analysis

We analyze the sensitivity of SMEL to five key hyperparameters: 1) the number of superpixels n_{sp}; 2) the anchor loss weight λ_a and the regional loss weight λ_r; 3) the PCA dimension d_{PCA}; and 4) the spectral grouping number g.

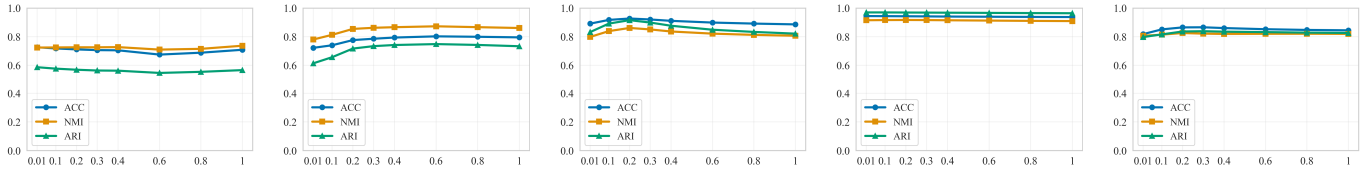
1) *n_{sp}:* The number of superpixels controls the spatial scale of regional evidence. When the number is too small, each superpixel may cover multiple semantic classes, causing regional averaging and regional targets to oversmooth local structures. When the number is too large, superpixels provide only limited region context and become close to local pixel blocks. As shown in Fig. 5, the performance varies smoothly within a proper range. This is because the proposed framework does not rely solely on superpixel information. Spectral evidence preserves pixel-level spectral information, while region-related evidence and regional targets serve as structural guidance. Therefore, moderate changes in the superpixel scale do not cause severe performance degradation.
2) *λ_a and λ_r:* The anchor loss controls the strength of global assignment stabilization. A very small λ_a may lead to assignment drift, while an overly large value may constrain the model by the initial soft anchor. The regional loss controls local structural refinement. A very small λ_r weakens the use of spatial continuity, whereas an overly large value may oversmooth boundaries and small objects. Figs. 6 and 7 show that the proposed method maintains comparable performance over a range of loss weights. This supports the complementary design of the two losses: the anchor loss preserves the global clustering coordinate, while the regional mixed-target loss improves local assignment consistency.
3) *d_{PCA}:* A small PCA dimension may discard useful spectral information, whereas a large one may retain redundant or noisy variations. As shown in Table VI, $d_{\text{PCA}} = 20$ provides the best overall balance across the five datasets and is therefore uniformly used in all experiments. Although a different dimension may benefit an individual scene, we uniformly use $d_{\text{PCA}} = 20$ for all datasets to avoid dataset-specific tuning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.



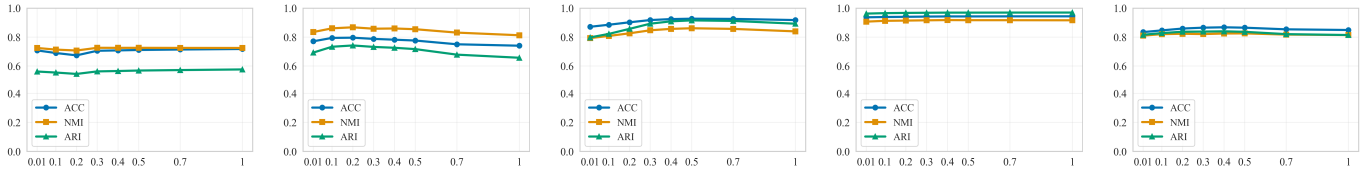
(a) PU (b) BO (c) TR (d) PC (e) LK

Fig. 5. Sensitivity analysis with respect to the number of superpixels on five benchmark datasets. (a) PU. (b) BO. (c) TR. (d) PC. (e) LK.



(a) (b) (c) (d) (e)

Fig. 6. Sensitivity analysis with respect to the anchor loss weight λ_a on five benchmark datasets. (a) PU. (b) BO. (c) TR. (d) PC. (e) LK.



(a) (b) (c) (d) (e)

Fig. 7. Sensitivity analysis with respect to the regional mixed-target loss weight λ_r on five benchmark datasets. (a) PU. (b) BO. (c) TR. (d) PC. (e) LK.

TABLE VI

SENSITIVITY ANALYSIS WITH RESPECT TO THE PCA DIMENSION d_{PCA}

Dataset	d_{PCA}=10			d_{PCA}=15			d_{PCA}=20			d_{PCA}=25			d_{PCA}=30		
	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
PU	0.650	0.641	0.494	0.655	0.644	0.494	0.709	0.717	0.570	0.660	0.649	0.504	0.664	0.654	0.514
BO	0.698	0.808	0.635	0.775	0.845	0.703	0.776	0.856	0.719	0.775	0.851	0.715	0.773	0.853	0.715
TR	0.929	0.867	0.922	0.929	0.867	0.922	0.930	0.868	0.923	0.930	0.869	0.924	0.931	0.870	0.925
PC	0.944	0.919	0.970	0.945	0.919	0.971	0.943	0.918	0.971	0.943	0.918	0.971	0.943	0.917	0.970
LK	0.880	0.832	0.853	0.864	0.824	0.833	0.870	0.827	0.841	0.872	0.829	0.844	0.884	0.836	0.859

4) *g:* The grouping number controls the spectral granularity: too few groups may obscure local variations, whereas too many may weaken aggregation. In this analysis, g denotes the coarse-grained grouping number, and the fine-grained descriptor uses $2g$ groups. Table VII shows that the default setting of $g = 8$, corresponding to 8 and 16 groups, provides the best overall performance across the five datasets. We therefore apply this fixed setting to all datasets, while finer dataset-specific adjustment may yield additional improvements in some cases.

Although extreme PCA dimensions, spectral grouping settings, superpixel scales, or loss weights may still harm performance, the proposed method is generally robust within common parameter ranges. This robustness improves its applicability to different hyperspectral scenes without requiring delicate hyperparameter tuning.

G. Model Complexity and Running Time Analysis

To evaluate the lightweight property of the proposed method, we first compare the total number of model parameters with representative deep hyperspectral clustering methods in Table VIII. The proposed method contains only 1.2K–5.7K parameters on the five datasets, with an average of 3.9K. This is substantially smaller than all compared methods. Even compared with lightweight methods, such as SLCGC and S2GCL, whose average parameter numbers are 18.0K and 19.8K, the proposed method still uses only about one fifth of their parameters. This indicates that the proposed framework does not rely on a large network capacity to obtain competitive clustering performance.

We further report the running time in Table IX. The proposed method is not the fastest one. Some lightweight methods, such as S2GCL and SSGCC, achieve lower running time by using simplified graph structures, region-level representations, or lightweight embedding modules. These designs can greatly reduce the computational cost, especially when the clustering process is mainly performed on a reduced graph or compact node representation. However, their performance may still depend on the reliability of the constructed graph, superpixel partition, or predefined structural relations.

In contrast, our method maintains pixel-level assignment learning while using superpixels mainly as regional evidence and training guidance. Therefore, its running time includes multi-evidence construction, low-rank evidence-specific

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
SENSITIVITY ANALYSIS WITH RESPECT TO THE SPECTRAL GROUPING NUMBER g

Dataset	g=4			g=6			g=8			g=10			g=12		
	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
PU	0.643	0.601	0.469	0.663	0.644	0.507	0.709	0.717	0.570	0.659	0.657	0.508	0.657	0.654	0.504
BO	0.702	0.806	0.618	0.728	0.815	0.645	0.776	0.856	0.719	0.714	0.822	0.639	0.632	0.767	0.563
TR	0.928	0.863	0.917	0.930	0.869	0.923	0.930	0.868	0.923	0.930	0.869	0.923	0.929	0.868	0.921
PC	0.938	0.909	0.970	0.943	0.918	0.970	0.943	0.918	0.971	0.898	0.886	0.961	0.943	0.919	0.970
LK	0.910	0.864	0.903	0.860	0.823	0.832	0.870	0.827	0.841	0.885	0.841	0.863	0.874	0.830	0.846

TABLE VIII
COMPARISON OF TOTAL MODEL PARAMETERS ON FIVE BENCHMARK DATASETS

Method	PU	BO	TR	PC	LK	Mean
NCSC [33]	298.3K	280.8K	309.8K	320.8K	348.3K	311.6K
DSCRLE [43]	212.0K	239.0K	186.6K	211.4K	316.9K	233.2K
S2GCL [44]	22.0K	22.0K	22.0K	22.0K	11.0K	19.8K
SSGCC [45]	39.75M	38.25M	39.03M	42.25M	39.03M	39.66M
SDST [13]	596.0K	609.7K	583.0K	595.7K	649.9K	606.8K
SLCGC [40]	11.0K	26.0K	21.0K	11.0K	21.0K	18.0K
LFGD [46]	417.9K	424.9K	2.38M	618.4K	936.0K	955.4K
EAGLE [12]	4.45M	1.13M	5.41M	4.45M	4.45M	3.98M
SSSC [47]	21.36M	54.79M	6.28M	79.93M	59.40M	44.35M
Ours	**1.2K**	**5.7K**	**2.8K**	**5.6K**	**4.1K**	**3.9K**

TABLE IX
RUNNING TIME COMPARISON ON FIVE BENCHMARK DATASETS. THE UNIT IS SECONDS AND THE MEAN VALUE IS COMPUTED OVER THE FIVE DATASETS UNDER THE SAME EVALUATION ENVIRONMENT

Method	PU	BO	TR	PC	LK	Mean
NCSC [33]	1166.0	82.6	845.8	3622.2	4212.7	1985.9
DSCRLE [43]	869.0	2137.3	444.3	4484.8	2305.8	2048.2
S2GCL [44]	6.4	10.0	4.6	18.8	17.5	11.5
SSGCC [45]	26.5	7.9	14.3	78.8	82.1	41.9
SDST [13]	10.8	253.8	10.2	159.6	21.0	91.1
SLCGC [40]	39.5	108.7	7.0	332.2	27.6	103.0
LFGD [46]	114.4	205.5	19.3	426.5	77.7	168.7
EAGLE [12]	119.9	645	775.2	665.9	92.1	459.6
SSSC [47]	3971.0	14284.1	645.2	20978.2	10163.9	10008.5
Ours	23.4	50.1	21.7	93.6	34.7	44.7

encoding, superpixel-refined teacher construction, anchor-region refinement, and final spatial readout. These operations introduce additional nonparametric cost, but they also help organize complementary evidence in a unified assignment space. As shown in Tables III and IX, the proposed method achieves a favorable tradeoff between clustering performance and practical efficiency. It is much faster than heavier methods on average, while maintaining a very small parameter scale and stronger overall clustering performance.

H. Limitations

Despite its effectiveness, the proposed method has two main limitations. The Stage-I warm-up may depend to some extent on the quality of the K-means partition, particularly for scenes with complex cluster structures. Such sensitivity is a common challenge in unsupervised HSI clustering and can be mitigated through multiple restarts or consensus clustering. In addition, the current evidence representations are constructed using predefined spectral–spatial organizations, and developing more adaptive evidence construction strategies may further enhance their flexibility across diverse scenes.

V. CONCLUSION

In this article, we proposed a SMEL framework for HSI clustering. To alleviate the unreliability of a single evidence source under spectral ambiguity and band redundancy, the proposed framework organizes complementary evidence from spectral characteristics, region context, and region prototype relations before assignment learning. Based on this structured evidence organization, a lightweight multi-evidence assignment learning module is developed to preserve evidence-specific information and establish semantic consensus among different evidence sources without relying on complex deep encoders or elaborate handcrafted priors. Extensive experiments on multiple benchmark hyperspectral datasets demonstrate that the proposed method achieves stable and competitive clustering performance with only a few thousand learnable parameters. These results validate the effectiveness of structured evidence organization for lightweight HSI clustering. In future work, we will investigate more adaptive evidence construction strategies and extend the framework to large-scale or multisource remote sensing clustering scenarios.

REFERENCES

[1] Z. Han, J. Yang, L. Gao, Z. Zeng, B. Zhang, and J. Chanussot, "Dual-branch subpixel-guided network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5521813.

[2] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[3] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[4] A. Bhargava, A. Sachdeva, K. Sharma, M. H. Alsharif, P. Uthansakul, and M. Uthansakul, "Hyperspectral imaging and its applications: A review," *Heliyon*, vol. 10, no. 12, Jun. 2024, Art. no. e33208.

[5] D. Hong et al., "Hyperspectral imaging," *Nature Rev. Methods Primers*, vol. 6, no. 1, p. 19, 2026.

[6] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[7] B. Han, Y. Jia, H. Liu, and J. Hou, "Irregular tensor low-rank representation for hyperspectral image representation," *IEEE Trans. Image Process.*, vol. 34, pp. 3239–3252, 2025.

[8] J. M. Amigo, H. Babamoradi, and S. Elcoroaristizabal, "Hyperspectral image analysis. A tutorial," *Analytica Chim. Acta*, vol. 896, pp. 34–51, Oct. 2015.

[9] G. Jiang, Y. Zhang, X. Wang, X. Jiang, and L. Zhang, "Structured anchor learning for large-scale hyperspectral image projected clustering," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 3, pp. 2328–2340, Mar. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.

[10] Y. Zhang, X. Wang, X. Jiang, L. Zhang, and B. Du, "Elastic graph fusion subspace clustering for large hyperspectral image," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 7, pp. 6300–6312, Jul. 2025.

[11] Q. Wang, Y. Miao, M. Chen, and Y. Yuan, "Spatial-spectral clustering with anchor graph for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542413.

[12] J. Qi, Y. Jia, H. Liu, and J. Hou, "Structural-spectral graph convolution with evidential edge learning for hyperspectral image clustering," *IEEE Trans. Image Process.*, vol. 34, pp. 7919–7929, 2025.

[13] F. Luo, Y. Liu, Y. Duan, T. Guo, L. Zhang, and B. Du, "SDST: Self-supervised double-structure transformer for hyperspectral images clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511614.

[14] Y. Yao, B. Peng, T. Qin, Y. Gu, N. Ling, and J. Lei, "Hypergraph contrastive learning for large-scale hyperspectral image clustering," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 7, pp. 7090–7100, Jul. 2025.

[15] M. Greenacre et al., "Principal component analysis," *Nature Rev. Methods Primers*, vol. 2, no. 1, p. 100, 2022.

[16] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[17] J. Qi, Y. Jia, H. Liu, and J. Hou, "Superpixel graph contrastive clustering with semantic-invariant augmentations for hyperspectral images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 11360–11372, Nov. 2024.

[18] Z. Han, J. Yang, L. Gao, Z. Zeng, B. Zhang, and J. Chanussot, "Subpixel spectral variability network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504014.

[19] H. Gao, R. Sheng, Y. Su, Z. Chen, S. Xu, and L. Gao, "Multiscale segmentation-guided fusion network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 34, pp. 6152–6167, 2025.

[20] W. Liu, S. Li, and X. Li, "Bi-temporal benefits: Progressive spectral–spatial–temporal feature extraction for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 35, pp. 5743–5757, 2026.

[21] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[22] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[23] R. Wang, F. Nie, and W. Yu, "Fast spectral clustering with anchor graph for large hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2003–2007, Nov. 2017.

[24] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[25] Y. Zhang, G. Jiang, Z. Cai, and Y. Zhou, "Bipartite graph-based projected clustering with local region guidance for hyperspectral imagery," *IEEE Trans. Multimedia*, vol. 26, pp. 9551–9563, 2024.

[26] Y. Zhang, S. Yan, L. Zhang, and B. Du, "Fast projected fuzzy clustering with anchor guidance for multimodal remote sensing imagery," *IEEE Trans. Image Process.*, vol. 33, pp. 4640–4653, 2024.

[27] U. Shaham, K. Stanton, H. Li, B. Nadler, R. Basri, and Y. Kluger, "SpectralNet: Spectral clustering using deep neural networks," 2018, *arXiv:1801.01587*.

[28] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.

[29] M. Zhang, T. Han, X. Qu, X. Gao, X. Liu, and S. Niu, "Masked superpixel contrastive subspace clustering network for unsupervised large-scale hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5520616.

[30] Y. Zhao, Z. Bi, P. Zhu, A. Yuan, and X. Li, "Deep spectral clustering with projected adaptive feature selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504712.

[31] S. Huang, H. Zeng, H. Chen, and H. Zhang, "Spatial and cluster structural prior-guided subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511115.

[32] K. Cui et al., "Superpixel-based and spatially regularized diffusion learning for unsupervised hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4405818.

[33] Y. Cai et al., "Superpixel contracted neighborhood contrastive subspace clustering network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530113.

[34] Z. Peng, H. Liu, Y. Jia, and J. Hou, "EGRC-Net: Embedding-induced graph refinement clustering network," *IEEE Trans. Image Process.*, vol. 32, pp. 6457–6468, 2023.

[35] Y. Ding et al., "Adaptive homophily clustering: Structure homophily graph learning with adaptive filter for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510113.

[36] X. Wang, Y. Zhang, and Y. Zhou, "Multimodal remote sensing image clustering with multiscale spectral–spatial anchor graphs," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405612.

[37] W. Liu, Y. Zhang, X. Wang, and L. Zhang, "Deep multi-level contrastive clustering for multi-modal remote sensing images," in *Proc. 33rd ACM Int. Conf. Multimedia*, Oct. 2025, pp. 1239–1247.

[38] Y. Zhang, K. Zheng, S. Yan, X. Wang, and Z. Cai, "Efficient tensorized multi-view anchor graph clustering with affinity propagation for remote sensing data," in *Proc. AAAI Conf. Artif. Intell.*, 2026, vol. 40, no. 34, pp. 28501–28509.

[39] F. Luo, Y. Liu, X. Gong, Z. Nan, and T. Guo, "EMVCC: Enhanced multi-view contrastive clustering for hyperspectral images," in *Proc. 32nd ACM Int. Conf. Multimedia*, New York, NY, USA, Oct. 2024, pp. 6288–6296.

[40] Y. Ding et al., "SLCGC: A lightweight self-supervised low-pass contrastive graph clustering network for hyperspectral images," *IEEE Trans. Multimedia*, vol. 27, pp. 8251–8262, 2025.

[41] Y. Chen, E. Li, A. Samat, J. Hu, W. Liu, and X. Li, "Spectral-structure enhanced multisubview contrastive clustering for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 10541–10554, 2026.

[42] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.

[43] Y. Zhao and X. Li, "Deep spectral clustering with regularized linear embedding for hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509311.

[44] A. Yang, M. Li, Y. Ding, X. Xiao, and Y. He, "An efficient and lightweight spectral–spatial feature graph contrastive learning framework for hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537714.

[45] R. Guan et al., "Spatial–spectral graph contrastive clustering with hard sample mining for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535216.

[46] A. Yang et al., "A robust low-pass filtering graph diffusion clustering framework for hyperspectral images," *Knowl.-Based Syst.*, vol. 324, Aug. 2025, Art. no. 113782.

[47] X. Li, N. Nadisic, S. Huang, and A. Pižurica, "Joint superpixel and self-representation learning for scalable hyperspectral image clustering," in *Proc. ICASSP - IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2026, pp. 546–550.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:12:03 UTC from IEEE Xplore. Restrictions apply.